

# AMAN PAL

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## PROFESSIONAL SUMMARY

Computer Science undergraduate (AI & ML) with hands-on experience building production-ready full-stack applications including collaborative IDEs, AI chatbots, and job portals. Proficient in React.js, Node.js and cloud technologies with a strong foundation in data structures and algorithms. Passionate about scalable, innovative solutions that solve real-world problems.

## EDUCATION

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| <b>B.Tech in Computer Science (AI &amp; ML)</b><br><i>Dr. A.P.J. Abdul Kalam Technical University, Lucknow</i> | <i>Sep 2023 – Present</i><br><i>Expected 2027</i> |
| <b>Intermediate (CBSE — Class XII)</b><br><i>Ram Prasad Bismil Memorial Public School</i>                      | <i>2023</i>                                       |

## TECHNICAL SKILLS

**Languages:** C++, C, JavaScript (ES6+), Java (Basic)  
**Frontend:** React.js, HTML5, CSS3, Tailwind CSS, Bootstrap, ShadCN UI, Responsive Design  
**Backend & Databases:** Node.js, Express.js, RESTful APIs, MongoDB, Supabase, Real-time Database Operations  
**Auth & Security:** Clerk, JWT, OAuth Integration  
**Real-time & AI:** CRDT, Yjs, WebSockets, Gemini API, Operational Transformation  
**Tools:** Git, VS Code Extension Development, Data Structures & Algorithms, Docker, CI/CD(Basic)

## EXPERIENCE

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| <b>Web Development Intern — CodTech Solutions</b><br><i>Remote</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>May 2025 – Jun 2025</i> |
| <ul style="list-style-type: none"><li>Developed and maintained full-stack web applications using React.js for dynamic UIs and Node.js/Express.js for robust backend services.</li><li>Designed and implemented RESTful APIs with MongoDB integration, ensuring efficient data management and seamless frontend-backend communication.</li><li>Optimized frontend components and backend logic to improve application load times and overall user experience.</li><li>Collaborated cross-functionally to deliver production-ready features across the complete software development lifecycle.</li></ul> |                            |

## PROJECTS

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| <b>CodeSphere — Collaborative Online IDE</b><br><i>React.js, TypeScript, CRDT, Yjs, Node.js, VS Code Extension</i>                                                                                                                                                                                                                                                                                                                                                                   | <i>2025</i> |
| <ul style="list-style-type: none"><li>Built a full-stack collaborative IDE for C++, Java, Python &amp; C with real-time multi-user editing via CRDT/Yjs for conflict-free synchronization.</li><li>Integrated Codeforces DSA problem sets in-IDE and implemented JWT/OAuth role-based auth to secure user sessions and workspaces.</li><li>Developed a companion VS Code Extension to sync local environments with CodeSphere workspaces, boosting developer productivity.</li></ul> |             |
| <b>Perceptron — AI Chatbot Application</b><br><i>React.js, Gemini API</i>                                                                                                                                                                                                                                                                                                                                                                                                            | <i>2025</i> |
| <ul style="list-style-type: none"><li>Engineered an intelligent chatbot using Google's Gemini API for context-aware real-time responses; optimized API calls and state management for efficient performance.</li></ul>                                                                                                                                                                                                                                                               |             |
| <b>Hirred — Job Portal Platform</b><br><i>React.js, Tailwind CSS, Supabase, ShadCN UI, Clerk</i>                                                                                                                                                                                                                                                                                                                                                                                     | <i>2025</i> |
| <ul style="list-style-type: none"><li>Built a full-featured job portal with role-based auth (Clerk), real-time database sync (Supabase), and recruiter/applicant profile management.</li></ul>                                                                                                                                                                                                                                                                                       |             |

## CERTIFICATIONS

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|----------------------------------------------------|-------------|
| Tata — GenAI Powered Data Analytics Job Simulation | <i>2025</i> |
| AWS — Solutions Architecture Job Simulation        | <i>2025</i> |